

DevStat — User Manual

Medical Statistics Software · Statistical analysis and visualisation of clinical data

This manual walks through every feature using real worked examples. All screenshots were generated from the included **sample medical dataset** (large clinical dataset with demographic, laboratory, treatment, and outcome variables).

1. Getting started

1.1 What DevStat does

DevStat is a desktop application for analysing clinical data — like a modern, easy-to-use version of SPSS. You load a dataset, choose a statistical test or chart, and get clear results you can view, export, and interpret.

- ▶ Import **CSV, Excel, SPSS (.sav), and Stata (.dta)** files
- ▶ View and edit data in a spreadsheet-style grid with SPSS-style variable metadata
- ▶ Run descriptive statistics, group comparisons, regression, survival, diagnostic tests, factor/reliability, cluster, and power analyses
- ▶ Create 27+ chart types and export results to PDF and images

1.2 Loading your data

Go to **Data** → **Data View** and click the **Upload** button. Choose your file. The data grid appears with rows and columns; the status bar shows the shape of the dataset (e.g. rows × columns).

For every example in this manual we use the sample medical dataset. Typical columns include:

Variable	Type	Meaning
age	Numeric	Age in years
bmi, systolic_bp, cholesterol, hemoglobin	Numeric	Continuous clinical measurements
gender	Categorical	Male / Female
treatment_group	Categorical	Drug A / Drug B / Placebo
outcome	Categorical	Improved / Not improved
survival_months	Numeric	Follow-up time for survival analysis
event_death	Binary	1 = event occurred, 0 = censored

2. Descriptive statistics

2.1 Descriptives

Analyze → Descriptive. Summarise one or more continuous variables (mean, standard deviation, minimum, maximum, skewness, kurtosis). Select variables and optionally a group to split results.

Descriptive Statistics

_columns	["age", "bmi", "cholesterol"]
_group_col	None
age	{ "n": 117, "missing": 3, "mean": 51.3584, "median": 51.0, "mode": 34.0, "std": 20.1393, "variance": 405.5917, "min": 19.0, "max": 98.0, "range": 79.0, "q1": 34.0, "q3": 69.0, "i": 24.275 }
bmi	{ "n": 116, "missing": 4, "mean": 27.769, "median": 27.3, "mode": 27.3, "std": 5.28, "variance": 27.878, "min": 10.9, "max": 45.0, "range": 34.1, "q1": 24.275, "q3": 30.8, "i": 24.275 }
cholesterol	{ "n": 117, "missing": 3, "mean": 5.2793, "median": 5.17, "mode": 4.17, "std": 1.41, "variance": 1.9882, "min": 2.26, "max": 12.5, "range": 10.24, "q1": 4.33, "q3": 6.15, "i": 24.275 }

Descriptive statistics table with N, mean, SD, min, max per variable.

2.2 Frequencies

Show the count and percentage of each category for a categorical variable (e.g. `smoking_status`). Optionally display a bar chart.

Frequencies - Smoking Status

column	smoking_status
n	118
missing	2
table	[{ "value": "Non-smoker", "count": 76, "percent": 64.41, "cumulative_percent": 64.41 }, { "value": "Ex-smoker", "count": 22, "percent": 18.64, "cumulative_percent": 83.05 }]
chart_type	frequencies

2.3 Crosstabs

Cross-tabulate two categorical variables and get the chi-square test of association. Useful for checking whether two categories are related (e.g. treatment group × outcome).

Crosstabs - Gender × Smoking Status

row	gender
col	smoking_status
n	115
table	[["", "Ex-smoker", "Non-smoker", "Smoker"], ["Female", 8, 27, 5], ["Male", 14, 46, 15]]
row_totals	{ "Female": 40, "Male": 75 }
col_totals	{ "Ex-smoker": 22, "Non-smoker": 73, "Smoker": 20 }
grand_total	115
expected	[["", "Ex-smoker", "Non-smoker", "Smoker"], ["Female", 7.6522, 25.3913, 6.9565], ["Male", 14.3478, 47.6087, 13.0435]]
percentages	{ "row": [["", "Ex-smoker", "Non-smoker", "Smoker"], ["Female", 20.0, 67.5, 12.5], ["Male", 18.67, 61.33,
chi2	1.0243
df	2
p_value	0.5992
cramers_v	0.0944
min_expected	6.9565

Contingency table with counts, percentages, and chi-square test.

2.4 Explore (normality)

Get descriptive statistics per group, plus normality tests (Shapiro–Wilk), Q–Q plots, and boxplots. Especially useful before choosing a test.

Explore - BMI

column	bmi
n	116
missing	4
descriptives	{ "mean": 27.769, "median": 27.3, "std": 5.28, "variance": 27.878, "min": 10.9, "max": 45.0, "range": 34.1, "q1": 24.275, "q3": 30.8, "iqr": 6.525 }
normality_tests	{ "shapiro_wilk": { "statistic": 0.967, "p_value": 0.00583696066628403, "normal": false }, "kolmogorov_smirnov": { "statistic": 0.0796, "p_value": 0.43228309419991673 } },
percentiles	{ "5": 20.975, "10": 21.85, "25": 24.275, "50": 27.3, "75": 30.8, "90": 33.95, "95": 36.7 }
outliers	{ "method": "Tukey's fences (1.5 \u00d7 IQR)", "lower_fence": 14.4875, "upper_fence": 40.5875, "n_outliers": 4, "pct_outliers": 3.45, "outliers_low": [10.9], "outliers_high": [] }
histogram	[{ "bin_start": 10.9, "bin_end": 13.5231, "count": 1 }, { "bin_start": 13.5231, "bin_end": 16.1462, "count": 0 }, { "bin_start": 16.1462, "bin_end": 18.7692, "count": 0 }]
interpretation	Normality diagnostics for 'bmi' (n = 116). Shapiro-Wilk test: W = 0.967, p = 0.0058 - the data deviates significantly from normality. Skewness = 0.5255 (z = 2.3483), significantly skewed. Kurtosis = 1.5512 (z = 3.4818), leptokurtic (significantly heavier tails than normal).

Group summaries, normality tests, and plots.

2.5 Means

Compare the mean of a continuous variable across groups, with confidence intervals.

Means - SBP by Treatment Group

eligible	False
blocked	True
requested_action	mean of treatment_group
action_type	descriptive
reason	Mean is meaningful only for numeric continuous data.
details	Variable 'treatment_group' is type 'nominal', which does not support 'mean'.
triggering_data_properties	["variable_type=nominal"]
suggested_alternatives	["Use frequency table (counts and percentages)."
alternative_ranked	{ "preferred": [], "acceptable": [], "advanced": [] }
inferred_data_role	{ }
help_terms	["mean", "continuous"]

3. Comparing means and groups

3.1 Independent-samples t-test

Analyze → Compare Means → Independent t-test. Compares the mean of a continuous variable between two unrelated groups (e.g. BMI by gender). Includes Levene's test for equal variances.

Independent t-test - BMI by Gender

test_name	Independent t-test
statistic	1.8549
df	105.2200
p_value	0.0664
effect_size	0.3246
effect_size_name	Cohen's d
effect_size_interpretation	small
levene	{ "statistic": 6.0509, "p_value": 0.015441314004376426, "equal_variances_assumed": false }
ci	{ "method": "mean difference 95% CI", "lower": -0.8673, "upper": 4.1442 }
descriptives	{ "Male": { "mean": 28.4257, "sd": 5.617, "ci_lower": 27.1243, "ci_upper": 29.727 }, "Female": { "mean": 26.7872, "sd": 3.7153, "ci_lower": 25.5828, "ci_upper": 28.0916 } }
n	{ "total": 113, "group_a": 74, "group_b": 39 }
interpretation	None

Group statistics and t-test results.

3.2 Paired t-test

Compare two measurements taken on the same subjects (e.g. systolic vs diastolic blood pressure).

Paired t-test - SBP vs DBP

test_name	Paired t-test
statistic	27.6747
df	113
p_value	0.0000
effect_size	2.5920
effect_size_name	Cohen's dz
effect_size_interpretation	large
ci	{ "method": "mean difference 95% CI", "lower": 44.2787, "upper": 51.1072 }
descriptives	{ "pre": { "mean": 128.8421, "sd": 15.8849, "ci_lower": 125.8946, "ci_upper": 131.7896 }, "post": { "mean": 81.1491, "sd": 10.2346, "ci_lower": 79.25, "ci_upper": 83.0477 } }
n	114
interpretation	None

Paired comparison of two related measurements.

3.3 One-way ANOVA

Compares the mean of a continuous variable across three or more groups (e.g. cholesterol by treatment group), with optional Tukey post-hoc tests.

One-way ANOVA - Cholesterol by Treatment

test_name	One-Way ANOVA
statistic	5.2202
statistic_name	F
df	{ "between": 2, "within": 111 }
p_value	0.0068
effect_size	0.0860
effect_size_name	Eta-squared (η^2)
effect_size_interpretation	medium
ss	{ "between": 19.6329, "within": 208.7341, "total": 228.367 }
ms	{ "between": 9.8165, "within": 1.8805 }
descriptives	{ "Drug A": { "mean": 5.23, "sd": 1.0746, "ci_lower": 4.918, "ci_upper": 5.542, "n": 48 }, "Placebo": { "mean": 6.149, "sd": 1.8771, "ci_lower": 5.2705, "ci_upper": 7.0275 } }
post_hoc	{ "method": "Tukey HSD", "comparisons": [{ "group_a": "Drug A", "group_b": "Drug B", "mean_difference": 0.2598, "p_value": 0.630017313827976, "significant": false }] }
n	114
k	3
interpretation	None

3.4 Non-parametric tests

Use these when data are not normally distributed or you have small samples:

- ▶ **Mann-Whitney U** — two independent groups (image below)
- ▶ **Wilcoxon signed-rank** — paired data
- ▶ **Kruskal-Wallis** — three or more groups
- ▶ **Chi-square / Fisher's exact** — categorical association
- ▶ **Friedman, McNemar, Binomial, Runs, Kolmogorov-Smirnov**

Mann-Whitney U - Pain Score by Gender

test_name	Mann-Whitney U
statistic	1348.0000
statistic_name	U
p_value	0.5767
effect_size	0.0639
effect_size_name	Rank-biserial correlation (r)
effect_size_interpretation	negligible
descriptives	{ "Male": { "n": 72, "median": 5.0, "iqr": 4.5, "mean_rank": 36.5 }, "Female": { "n": 40, "median": 6.0, "iqr": 5.25, "mean_rank": 20.5 } }
n	{ "total": 112, "group_a": 72, "group_b": 40 }
interpretation	None

Mann-Whitney U test output.

4. Correlation and regression

4.1 Correlation

Analyze → Correlation. Compute a matrix of correlations between continuous variables (Pearson or Spearman). Optionally partial correlation controlling for a third variable.

Pearson Correlation Matrix

matrix	[{ "age": 1.0, "bmi": 0.2624, "systolic_bp": 0.2835, "cholesterol": -0.0563 }, { "age": 0.2624, "bmi": 1.0, "systolic_bp": 0.2791, "cholesterol": 0.0555 }, {
p_values	[{ "age": 1.0, "bmi": 0.0063, "systolic_bp": 0.0031, "cholesterol": 0.5646 }, { "age": 0.0063, "bmi": 1.0, "systolic_bp": 0.0036, "cholesterol": 0.57 }, {
n	[{ "age": 107, "bmi": 107, "systolic_bp": 107, "cholesterol": 107 }, { "age": 107, "bmi": 107, "systolic_bp": 107, "cholesterol": 107 }, { "age": 107,
columns	["age", "bmi", "systolic_bp", "cholesterol"]
method	pearson
n_complete	107
interpretation	A Pearson correlation matrix was computed for 4 variables (age, bmi, systolic_bp, cholesterol). No notable correlations ($ r \geq 0.3$) were observed among the variables.

Pearson correlation matrix with r and p -values.

4.2 Linear regression

Model a continuous outcome from one or more predictors (e.g. systolic BP from age, BMI, cholesterol). Reports R^2 , the ANOVA table, coefficients, and p -values.

Linear Regression - SBP ~ Age + BMI + Cholesterol

coefficients	[{ "name": "const", "coef": 105.6935, "se": 9.6059, "t": 11.003, "p": 0.0, "p_label": { "value": 4.380318591268402e-19, "label": "p < 0.001", "sig": "****" }
model_summary	{ "r_squared": 0.1337, "adj_r_squared": 0.1084, "f_stat": 5.2969, "f_p": 0.002, "f_p_label": { "value": 0.002, "label": "p = 0.002", "sig": "****" }, "aic": 887.0845, "bic": }
anova_table	{ "anova_table": [], "note": "ANOVA table could not be computed (likely single-predictor model handled internally)."} }
residuals_notes	{ "normality": { "test": "Shapiro-Wilk", "statistic": 0.9658, "p": 0.0074, "p_label": { "value": 0.0074, "label": "p = 0.007", "sig": "****" }, "interpretatio
interpretation	A linear regression was performed with systolic_bp as the dependent variable and 3 predictor(s) (age, bmi, cholesterol). The overall model was statistically significant, $F(3, 103) = 5.2969$, $p = 0.002$. The model explained 13.4% of the variance in the outcome (adjusted $R^2 = 0.1084$). Significant predictors: • age: $B = 0.1729$ ($SE = 0.0753$), $t = 2.2967$, $p = 0.024$, $\beta = 0.2189$ • bmi: $B = 0.6988$ ($SE = 0.2937$), $t = 2.3798$, $p = 0.019$, $\beta = 0.2268$

Model summary, coefficients, and significance.

4.3 Logistic regression

Model a binary outcome (e.g. diabetes yes/no) from predictors. Reports odds ratios and Wald statistics.

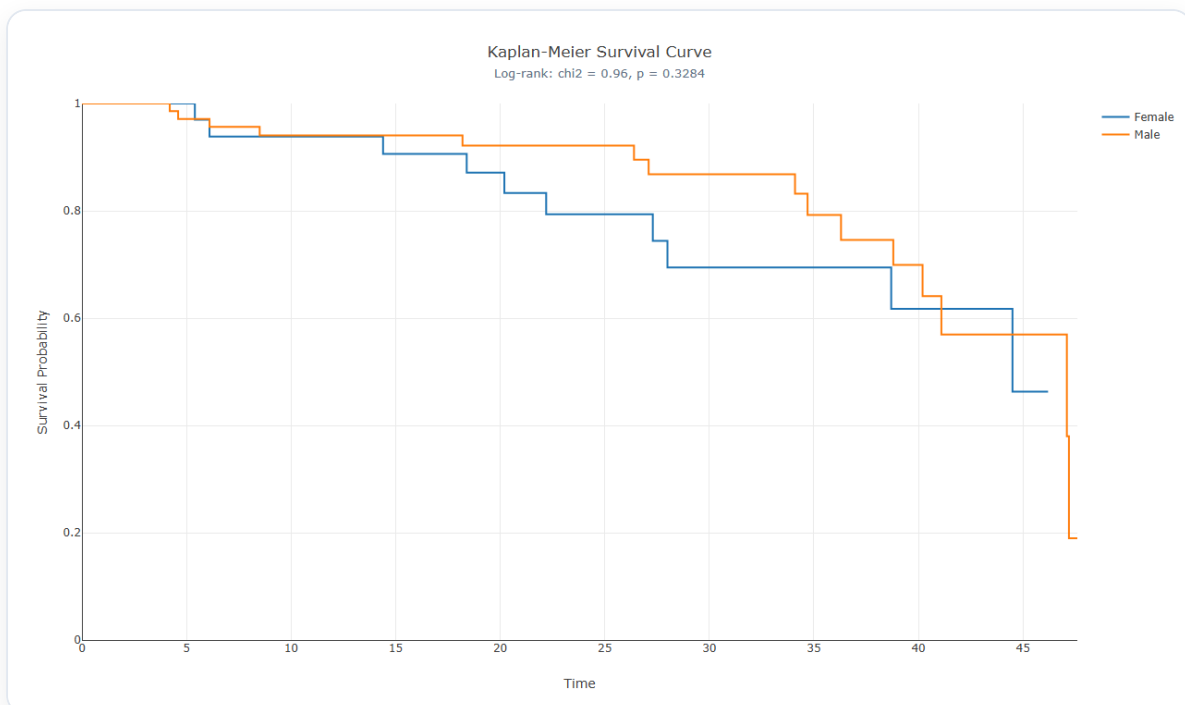
Logistic Regression - Diabetes ~ Age + BMI + Cholesterol

coefficients	[{ "name": "const", "coef": 0.946, "se": 1.7699, "z": 0.5345, "p": 0.593, "p_label": { "value": 0.593, "label": "p = 0.593", "sig": "ns" }, "or": 2 }
model_summary	{ "pseudo_r_squared": 0.0326, "log_likelihood": -49.1524, "null_log_likelihood": -50.809, "lr_stat": 3.3131, "lr_p": 0.3458, "lr_p_label": { "value": 0.3458, "label": "p = 0.346",
classification_table	{ "tp": 0, "fp": 0, "fn": 19, "tn": 92, "accuracy": 0.8288, "sensitivity": 0.0, "specificity": 1.0, "ppv": 0.0, "npv": 0.8288, "threshold": 0.5 }
interpretation	A binary logistic regression was performed with diabetes_bin as the outcome and 3 predictor(s) (age, bmi, cholesterol). The overall model was not statistically significant, $\chi^2(3) = 3.3131$, $p = 0.346$. Nagelkerke pseudo- $R^2 = 0.033$, indicating that the model explains approximately 3.3% of the variance in the outcome. The model correctly classified 82.9% of cases (sensitivity = 0.0%, specificity = 100.0%). No individual predictors reached statistical significance ($p < 0.05$).

5. Survival analysis

5.1 Kaplan-Meier

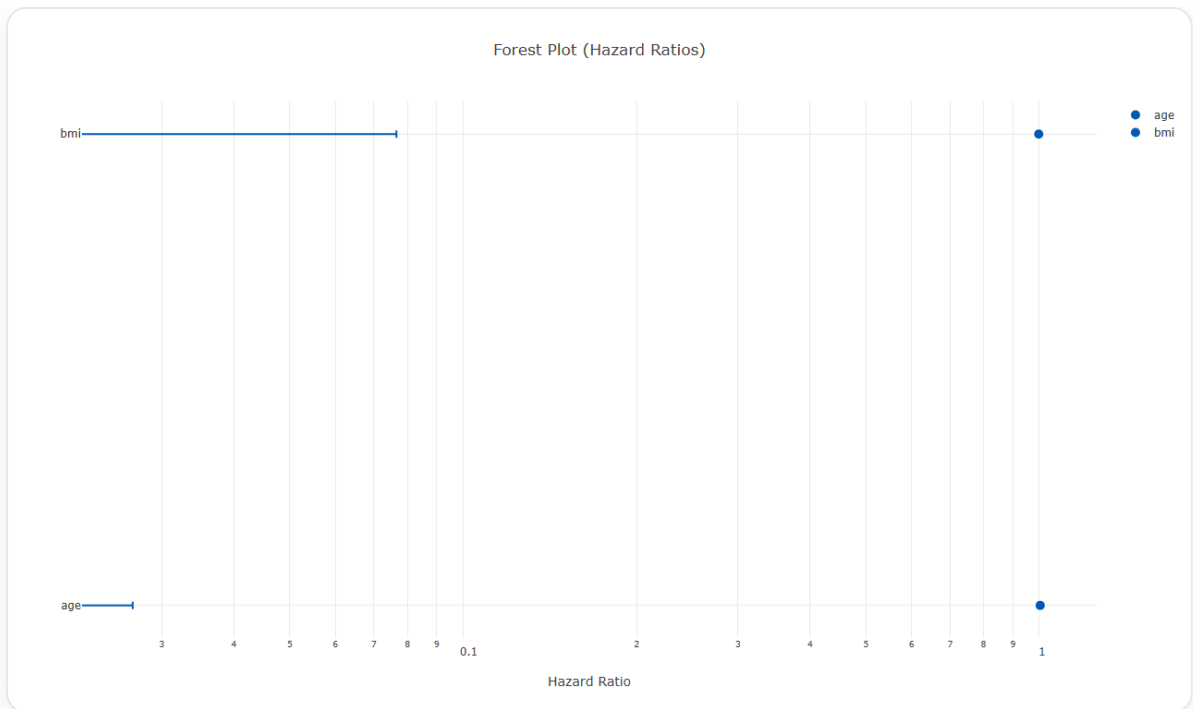
Survival → **Kaplan-Meier**. Time-to-event analysis with a time variable and a status (event) column. Produces the survival curve, median survival, and log-rank test.



Kaplan-Meier survival curve with 95% confidence band.

5.2 Cox regression

Model the hazard of an event from covariates. Reports hazard ratios and confidence intervals.



Cox proportional hazards output.

6. Diagnostic tests and ROC

6.1 Diagnostic test

Compare a test against a gold standard. Produces the 2×2 table (TP, FP, FN, TN) plus sensitivity, specificity, PPV, NPV, and accuracy.

Diagnostic Test - Pain Score vs Event

tp	5
fp	4
fn	21
tn	80
n	110
sensitivity	0.1923
specificity	0.9524
ppv	0.5556
npv	0.7921
positive_likelihood_ratio	4.0385
negative_likelihood_ratio	0.8481
prevalence	0.2364
accuracy	0.7727
f1_score	0.2857
youden_index	0.1447
mcc	0.2243
test_positive_code	1
gold_positive_code	1
interpretation	A diagnostic test evaluation was performed on 110 subjects using a 2x2 table. The test demonstrates low sensitivity (19.2%) and high specificity (95.2%). Positive predictive value is 55.6%, negative predictive value is 79.2%. The positive likelihood ratio is 4.04, providing limited diagnostic evidence. Overall accuracy: 77.3%, with a Youden index of 0.145 and Matthews correlation coefficient (MCC) of 0.224.

2x2 test table with diagnostic metrics.

6.2 ROC curve

Plot the receiver operating characteristic curve for a continuous test variable against a binary outcome, with the area under the curve (AUC).

ROC Analysis - Age predicting Event

roc_coordinates	[{ "fpr": 0.0, "tpr": 0.0, "threshold": null }, { "fpr": 0.0115, "tpr": 0.0, "threshold": 98.0 }, { "fpr": 0.023, "tpr": 0.04, "threshold": 85.0 }, { "fpr": 0.0345, "tpr": 0.08, "threshold": 80.0 }, { "fpr": 0.046, "tpr": 0.12, "threshold": 75.0 }, { "fpr": 0.0575, "tpr": 0.16, "threshold": 70.0 }, { "fpr": 0.069, "tpr": 0.2, "threshold": 65.0 }, { "fpr": 0.0805, "tpr": 0.24, "threshold": 60.0 }, { "fpr": 0.092, "tpr": 0.28, "threshold": 55.0 }, { "fpr": 0.1035, "tpr": 0.32, "threshold": 50.0 }, { "fpr": 0.115, "tpr": 0.36, "threshold": 45.0 }, { "fpr": 0.1265, "tpr": 0.4, "threshold": 40.0 }, { "fpr": 0.138, "tpr": 0.44, "threshold": 35.0 }, { "fpr": 0.1495, "tpr": 0.48, "threshold": 30.0 }, { "fpr": 0.161, "tpr": 0.52, "threshold": 25.0 }, { "fpr": 0.1725, "tpr": 0.56, "threshold": 20.0 }, { "fpr": 0.184, "tpr": 0.6, "threshold": 15.0 }, { "fpr": 0.1955, "tpr": 0.64, "threshold": 10.0 }, { "fpr": 0.207, "tpr": 0.68, "threshold": 5.0 }, { "fpr": 0.2185, "tpr": 0.72, "threshold": 0.0 }, { "fpr": 0.23, "tpr": 0.76, "threshold": -5.0 }, { "fpr": 0.2415, "tpr": 0.8, "threshold": -10.0 }, { "fpr": 0.253, "tpr": 0.84, "threshold": -15.0 }, { "fpr": 0.2645, "tpr": 0.88, "threshold": -20.0 }, { "fpr": 0.276, "tpr": 0.92, "threshold": -25.0 }, { "fpr": 0.2875, "tpr": 0.96, "threshold": -30.0 }, { "fpr": 0.299, "tpr": 1.0, "threshold": -35.0 }, { "fpr": 0.3105, "tpr": 1.0, "threshold": -40.0 }, { "fpr": 0.322, "tpr": 1.0, "threshold": -45.0 }, { "fpr": 0.3335, "tpr": 1.0, "threshold": -50.0 }, { "fpr": 0.345, "tpr": 1.0, "threshold": -55.0 }, { "fpr": 0.3565, "tpr": 1.0, "threshold": -60.0 }, { "fpr": 0.368, "tpr": 1.0, "threshold": -65.0 }, { "fpr": 0.3795, "tpr": 1.0, "threshold": -70.0 }, { "fpr": 0.391, "tpr": 1.0, "threshold": -75.0 }, { "fpr": 0.4025, "tpr": 1.0, "threshold": -80.0 }, { "fpr": 0.414, "tpr": 1.0, "threshold": -85.0 }, { "fpr": 0.4255, "tpr": 1.0, "threshold": -90.0 }, { "fpr": 0.437, "tpr": 1.0, "threshold": -95.0 }, { "fpr": 0.4485, "tpr": 1.0, "threshold": -100.0 }, { "fpr": 0.46, "tpr": 1.0, "threshold": -105.0 }, { "fpr": 0.4715, "tpr": 1.0, "threshold": -110.0 }, { "fpr": 0.483, "tpr": 1.0, "threshold": -115.0 }, { "fpr": 0.4945, "tpr": 1.0, "threshold": -120.0 }, { "fpr": 0.506, "tpr": 1.0, "threshold": -125.0 }, { "fpr": 0.5175, "tpr": 1.0, "threshold": -130.0 }, { "fpr": 0.529, "tpr": 1.0, "threshold": -135.0 }, { "fpr": 0.5405, "tpr": 1.0, "threshold": -140.0 }, { "fpr": 0.552, "tpr": 1.0, "threshold": -145.0 }, { "fpr": 0.5635, "tpr": 1.0, "threshold": -150.0 }, { "fpr": 0.575, "tpr": 1.0, "threshold": -155.0 }, { "fpr": 0.5865, "tpr": 1.0, "threshold": -160.0 }, { "fpr": 0.598, "tpr": 1.0, "threshold": -165.0 }, { "fpr": 0.6095, "tpr": 1.0, "threshold": -170.0 }, { "fpr": 0.621, "tpr": 1.0, "threshold": -175.0 }, { "fpr": 0.6325, "tpr": 1.0, "threshold": -180.0 }, { "fpr": 0.644, "tpr": 1.0, "threshold": -185.0 }, { "fpr": 0.6555, "tpr": 1.0, "threshold": -190.0 }, { "fpr": 0.667, "tpr": 1.0, "threshold": -195.0 }, { "fpr": 0.6785, "tpr": 1.0, "threshold": -200.0 }, { "fpr": 0.69, "tpr": 1.0, "threshold": -205.0 }, { "fpr": 0.7015, "tpr": 1.0, "threshold": -210.0 }, { "fpr": 0.713, "tpr": 1.0, "threshold": -215.0 }, { "fpr": 0.7245, "tpr": 1.0, "threshold": -220.0 }, { "fpr": 0.736, "tpr": 1.0, "threshold": -225.0 }, { "fpr": 0.7475, "tpr": 1.0, "threshold": -230.0 }, { "fpr": 0.759, "tpr": 1.0, "threshold": -235.0 }, { "fpr": 0.7705, "tpr": 1.0, "threshold": -240.0 }, { "fpr": 0.782, "tpr": 1.0, "threshold": -245.0 }, { "fpr": 0.7935, "tpr": 1.0, "threshold": -250.0 }, { "fpr": 0.805, "tpr": 1.0, "threshold": -255.0 }, { "fpr": 0.8165, "tpr": 1.0, "threshold": -260.0 }, { "fpr": 0.828, "tpr": 1.0, "threshold": -265.0 }, { "fpr": 0.8395, "tpr": 1.0, "threshold": -270.0 }, { "fpr": 0.851, "tpr": 1.0, "threshold": -275.0 }, { "fpr": 0.8625, "tpr": 1.0, "threshold": -280.0 }, { "fpr": 0.874, "tpr": 1.0, "threshold": -285.0 }, { "fpr": 0.8855, "tpr": 1.0, "threshold": -290.0 }, { "fpr": 0.897, "tpr": 1.0, "threshold": -295.0 }, { "fpr": 0.9085, "tpr": 1.0, "threshold": -300.0 }, { "fpr": 0.92, "tpr": 1.0, "threshold": -305.0 }, { "fpr": 0.9315, "tpr": 1.0, "threshold": -310.0 }, { "fpr": 0.943, "tpr": 1.0, "threshold": -315.0 }, { "fpr": 0.9545, "tpr": 1.0, "threshold": -320.0 }, { "fpr": 0.966, "tpr": 1.0, "threshold": -325.0 }, { "fpr": 0.9775, "tpr": 1.0, "threshold": -330.0 }, { "fpr": 0.989, "tpr": 1.0, "threshold": -335.0 }, { "fpr": 1.0, "tpr": 1.0, "threshold": -340.0 }]
auc	0.5395
optimal_cutoff	31.0000
sensitivity_at_cutoff	0.9200
specificity_at_cutoff	0.2414
youden_index	0.1614
n	112
n_positive	25
n_negative	87
test_column	age
gold_column	event_occurred
interpretation	ROC analysis yielded an AUC of 0.5395, which is considered failing / no better than chance discriminatory ability. The optimal cutoff value (by Youden index) is 31.0000, with sensitivity = 92.0% and specificity = 24.1%. Analysis performed on 112 observations (25 positive, 87 negative).

7. Factor analysis, reliability, and clustering

7.1 Factor analysis

Identify underlying factors among many variables. Reports KMO, Bartlett's test, factor loadings, and variance explained.

Factor Analysis (2 factors)

loadings	[{ "variable": "age", "factor_1": 0.3827, "factor_2": -0.2286 }, { "variable": "bmi", "factor_1": 0.7952, "factor_2": 0.2031 }, { "variable": "systolic_bp", "factor_1": 0.6047, "factor_2": 0.6047 }
eigenvalues	[1.6415009177234468, 1.1652871330997912, 1.0377192723923825, 0.845279420091366, 0.7054370573808922, 0.6047761993121218]
variance_explained	[{ "factor": 1, "eigenvalue": 1.6415, "variance_pct": 27.36, "cumulative_pct": 27.36 }, { "factor": 2, "eigenvalue": 1.1653, "variance_pct": 19.42, "cumulative_pct": 46.78 }]
communalities	[{ "variable": "age", "communality": 0.1987 }, { "variable": "bmi", "communality": 0.6736 }, { "variable": "systolic_bp", "communality": 0.3458 }]
rotation	varimax
n_factors	2
n_variables	6
n_observations	101
kmo	{ "model": 0.573, "interpretation": "miserable" }
bartlett	{ "chi2": 32.4846, "df": 15, "p_value": 0.005527499852318546 }
interpretation	An exploratory factor analysis with varimax rotation was performed on 6 variables using principal axis factoring. KMO measure of sampling adequacy was 0.573 (miserable), and Bartlett's test of sphericity was significant (p = 0.0055), supporting the suitability of the data for factor analysis. The 2-factor solution explained 100.00% of the total variance. Factor 1 loaded on: bmi (+0.80), systolic_bp (+0.47).

KMO/Bartlett's test and rotated factor loadings.

7.2 Reliability (Cronbach's alpha)

Assess internal consistency of a scale using Cronbach's alpha.

Reliability Analysis (Cronbach's Alpha)

alpha	-0.0398
alpha_ci	[-0.424, 0.254]
n_items	3
n_observations	112
item_stats	[{ "item": "anxiety_score", "mean": 1.4732, "sd": 1.1544, "item_total_correlation": -0.1523, "alpha_if_deleted": -0.0398 }, { "item": "depression_score", "mean": 1.2
interpretation	Unacceptable internal consistency ($\alpha < 0.5$). The items do not reliably measure a common construct.

Cronbach's alpha and item statistics.

7.3 Cluster analysis

Group observations by similarity (k-means or hierarchical), producing cluster sizes, centroids, and a silhouette score.

K-Means Cluster Analysis (3 clusters)

method	kmeans
n_clusters	3
n_observations	107
n_variables	4
cluster_sizes	{ "0": 26, "1": 32, "2": 49 }
silhouette_score	0.2243
inertia	283.9192
columns	["age", "bmi", "systolic_bp", "cholesterol"]
cluster_centers	[{ "age": 63.0385, "bmi": 24.1346, "systolic_bp": 135.7308, "cholesterol": 6.08 }, { "age": 68.5312, "bmi": 32.4125, "systolic_bp": 133.7812, "cholesterol": 4.79
interpretation	A kmeans cluster analysis was performed on 107 observations with 4 variables, producing 3 clusters. Cluster sizes: Cluster 0: n=26, Cluster 1: n=32, Cluster 2: n=49. The silhouette score is 0.224, suggesting weak cluster structure.

8. Power analysis

Determine the sample size needed to detect a given effect (t-test, ANOVA, or correlation) at a chosen power and alpha.

Power Analysis - Independent t-test

test	Independent t-test (two-sided)
effect_size	0.5000
effect_size_measure	Cohen's d
alpha	0.0500
power	0.8000
n_total	34
n_per_group	34
interpretation	Required total sample size: 34 (17 per group for equal groups).

Required sample size for a t-test.

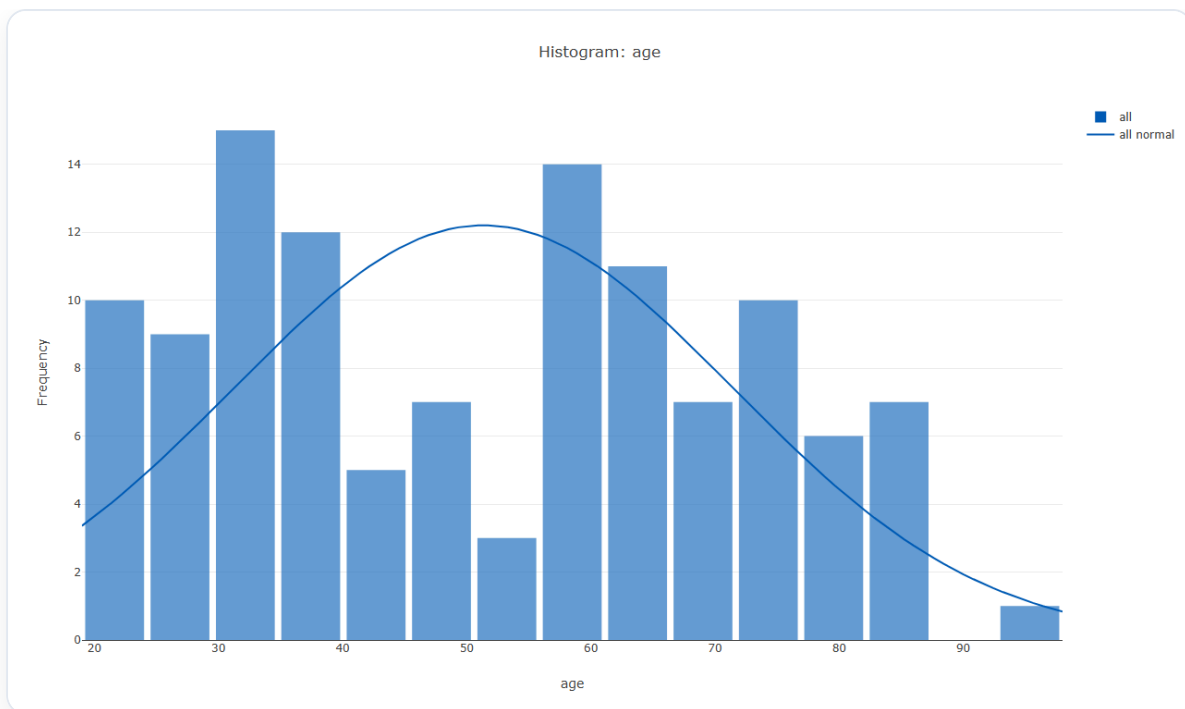
Power Analysis - ANOVA (3 groups)

test	One-way ANOVA (3 groups)
effect_size	0.2500
effect_size_measure	Cohen's f
alpha	0.0500
power	0.8000
n_groups	3
n_total	158
n_per_group	53
interpretation	Required total sample size: 158 (~53 per group).

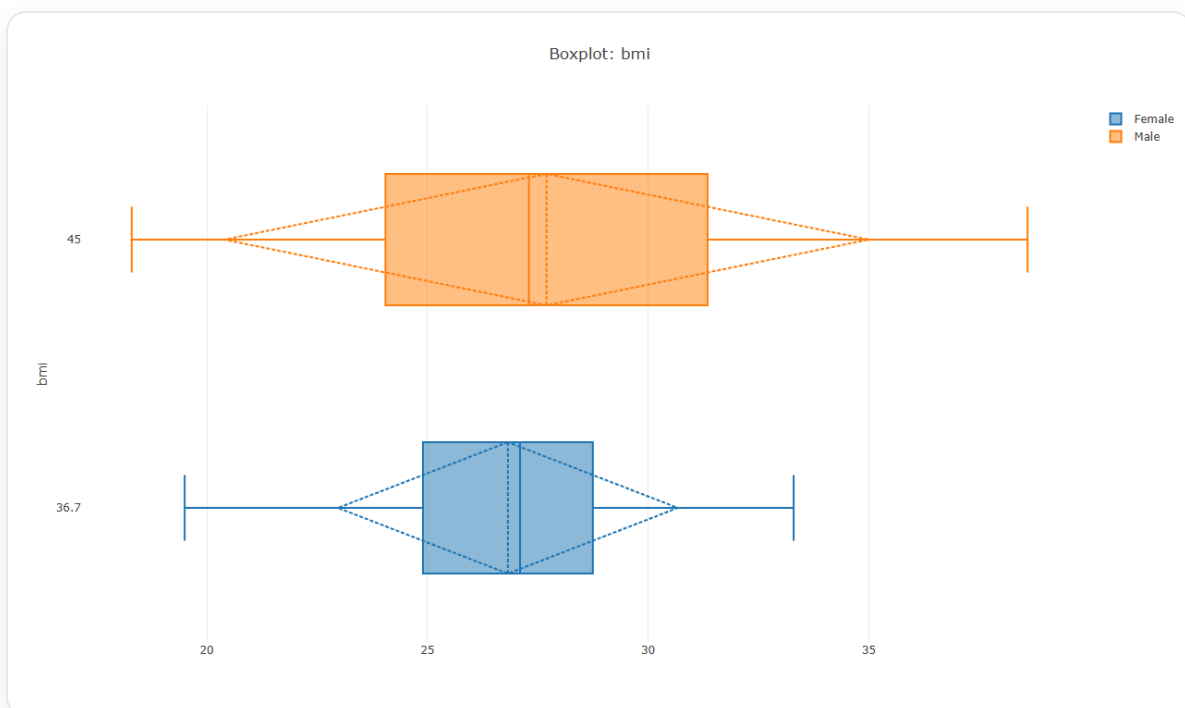
Required sample size for one-way ANOVA.

9. Graphs and charts

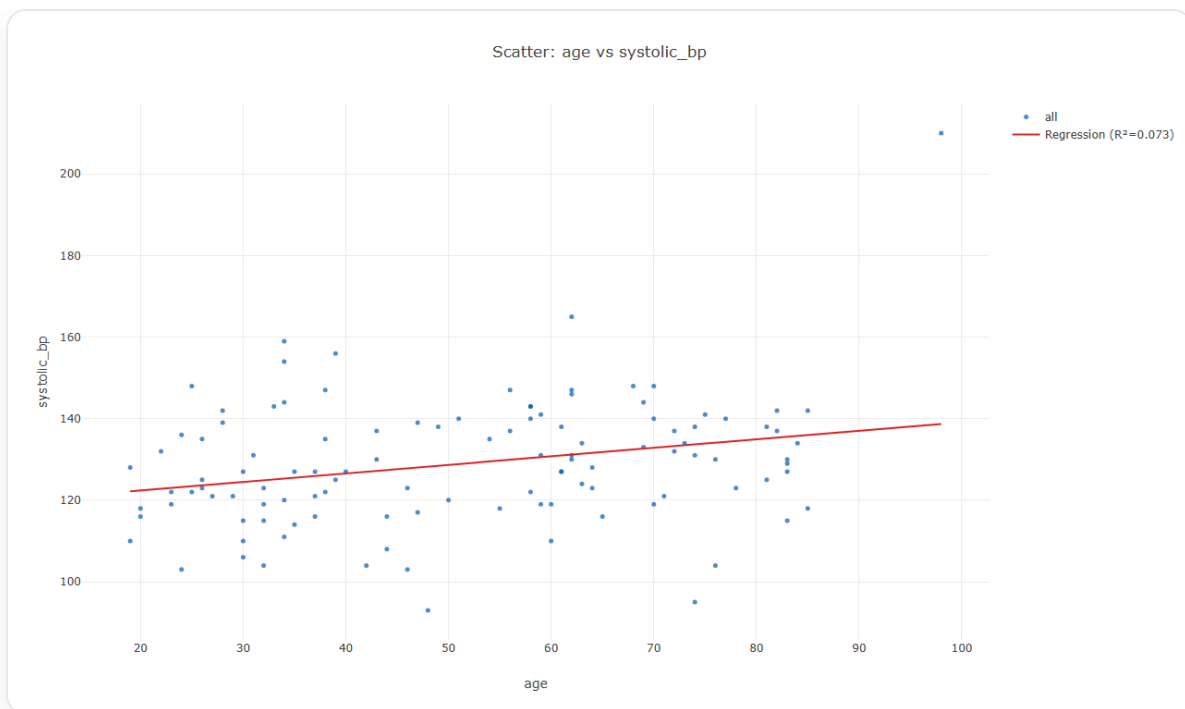
The **Graphs** menu offers a gallery of chart types. All are interactive: hover for values, zoom, and export to PNG.



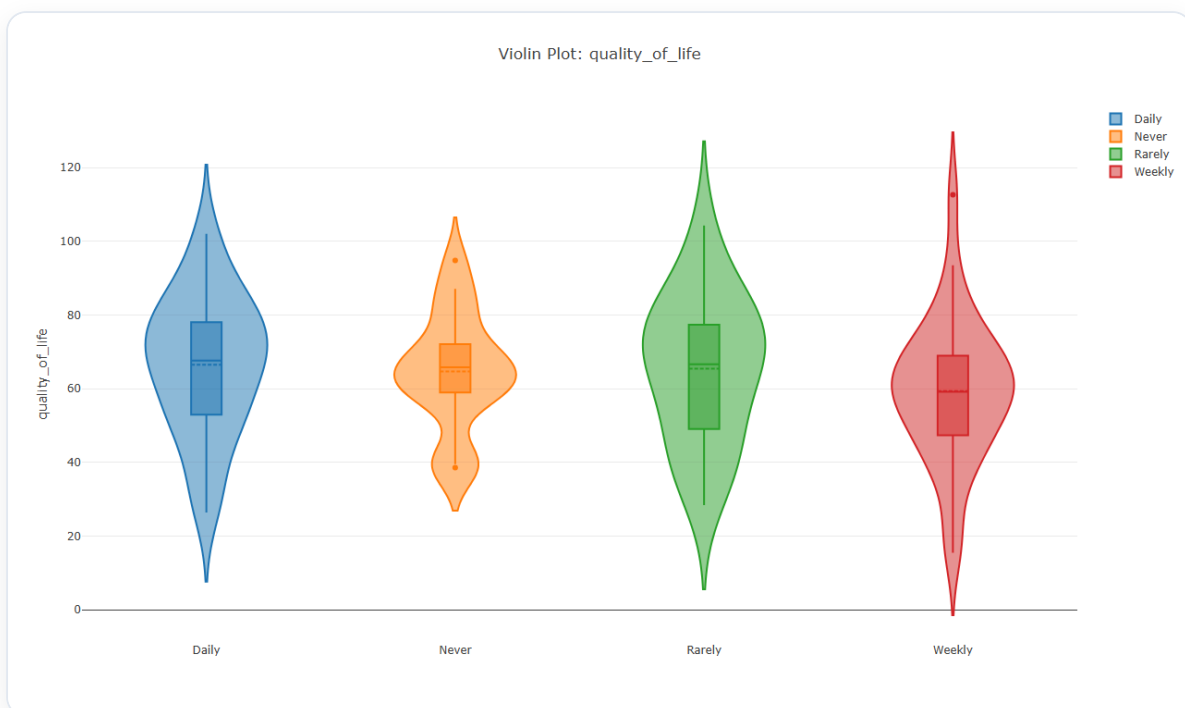
Histogram of a continuous variable, with a normal reference curve.



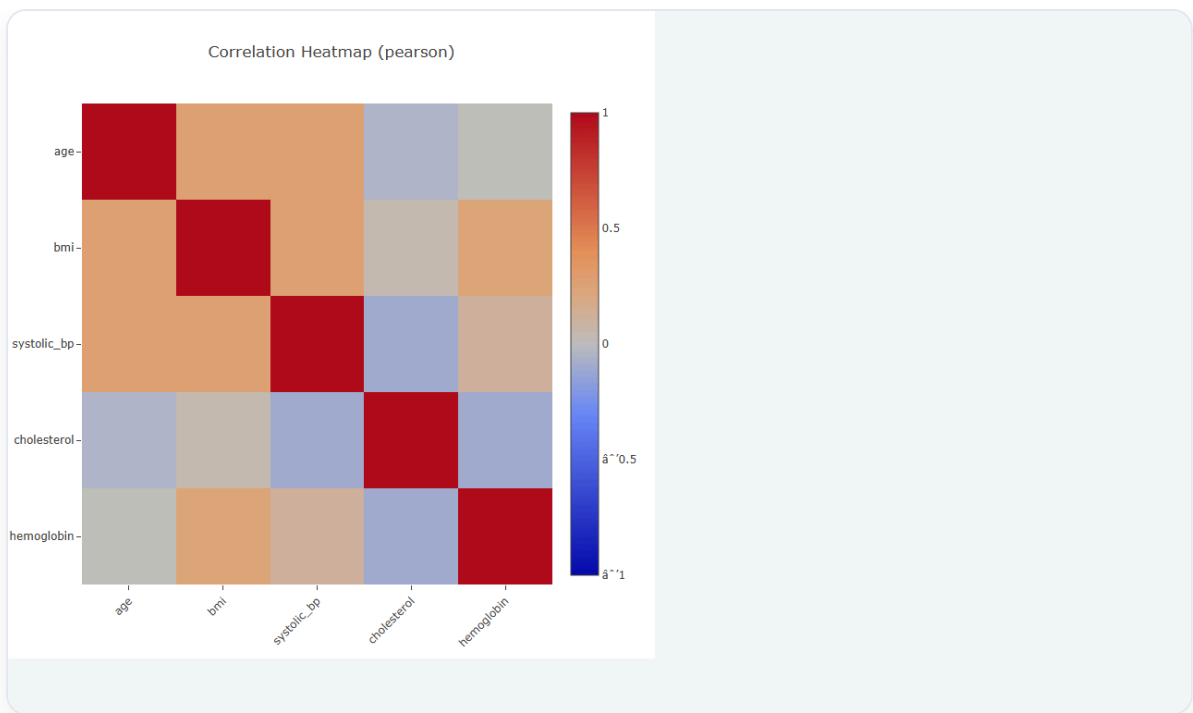
Boxplot of a variable (optionally by group).



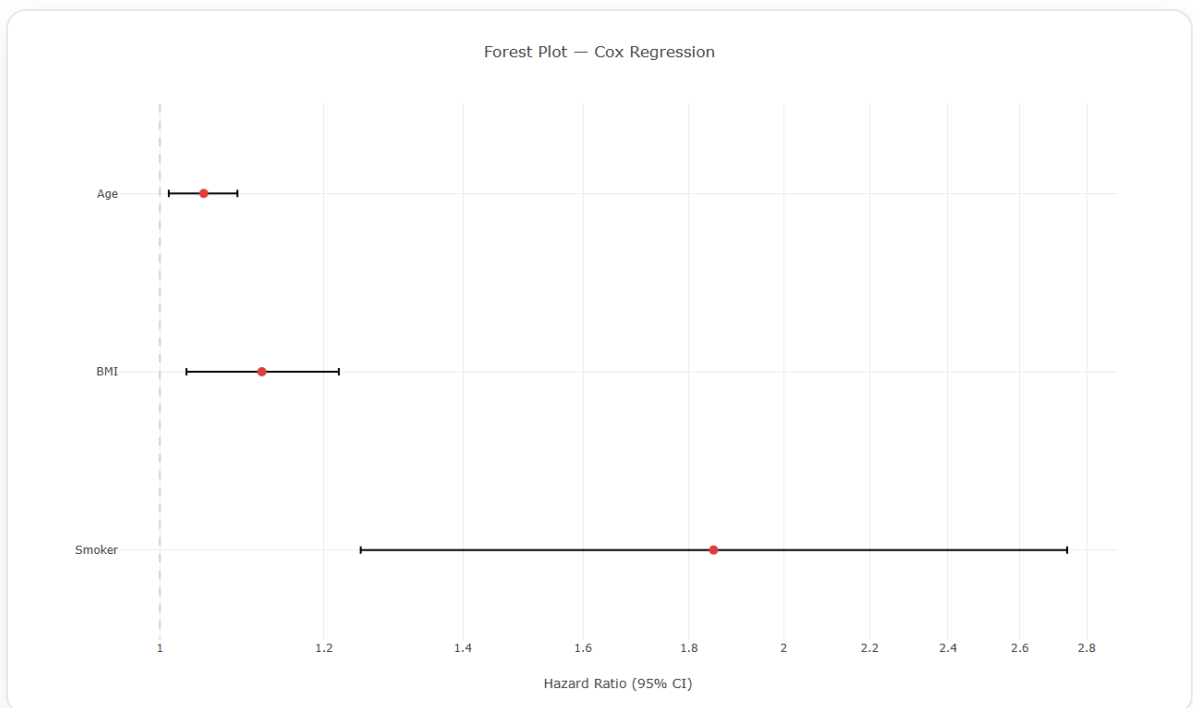
Scatter plot with regression fit line.



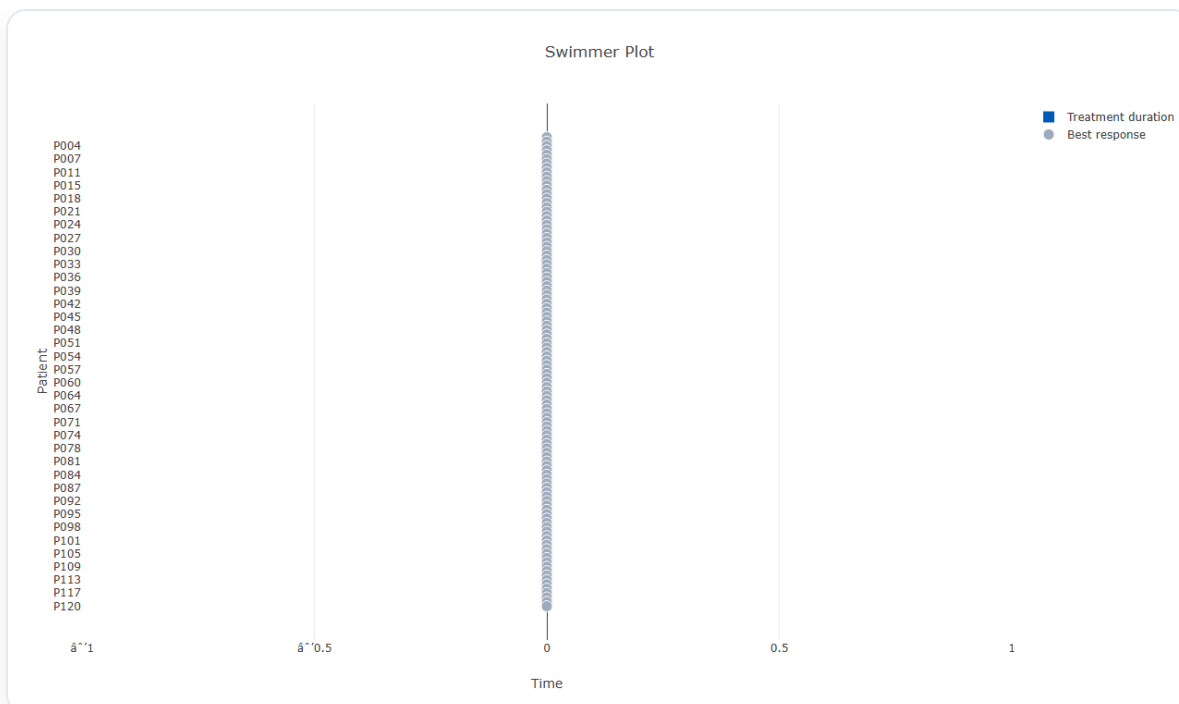
Violin plot showing distribution shape by group.



Correlation heatmap.



Forest plot (effect sizes with confidence intervals).



Swimmer plot (treatment timelines).

10. Output, export, and saving

- **Output** — a running list of every analysis performed, organised by category. Select items to compare side by side.
- **Export** — results as PDF, charts as PNG, and the dataset as CSV, Excel, SPSS (.sav), or Stata (.dta).
- **Save / Load project** — save your current dataset and analysis view to a .devstat file for later.

11. A short guide to interpreting output

Reading a p-value

A p-value (< 0.05) suggests the result is statistically significant — unlikely to be due to chance. It does not tell you how large the effect is; look at the estimate (mean difference, hazard ratio, correlation) and its confidence interval for practical importance.

Choosing the right test

Use the **Wizard** or **Suggest Test** feature: answer a few questions about your goal, variable types, and design, and DevStat recommends an appropriate test. Hover over any test or chart card in the gallery to see what it is for and whether your current data are suitable.

Common pitfalls

- ▶ **Normality** — use the Explore feature (Shapiro–Wilk). If data are not normal, prefer a non-parametric test.
- ▶ **Multiple testing** — correct for many comparisons (e.g. Tukey post-hoc after ANOVA).
- ▶ **Confidence intervals** — a wide interval means an imprecise estimate (often a small sample or few events, e.g. high censoring in survival data).

Disclaimer: DevStat is a statistical tool for research and analysis. Results should be interpreted by a qualified statistician or clinician in context of the study design. It does not provide medical advice.